CONFIDENTIAL INDUSTRIAL MAINTENANCE REPORT

Hydraulic Turbine HT-500 — Unit 17

Report ID	HT500-IR-017-2026	Revision	1.0
Facility	Western Hydro Operations — Demo Plant	Inspection Date	28 Aug 2026
Equipment	Hydraulic Turbine HT-500	Inspection Type	Routine + Condition Assessment
Prepared By	Maintenance Engineering Cell	Status	REQUIRES REVIEW

1. Executive Summary

A scheduled inspection of Hydraulic Turbine HT-500, Unit 17, identified multiple operating parameters outside the demonstration plant's recommended ranges. Elevated bearing temperature, hydraulic pressure and vibration were recorded during the loaded operating condition. Lubricating-oil level was also below the preferred operating band.

The observed conditions do not indicate an immediate catastrophic failure, but the combination of deviations warrants controlled maintenance action and engineering review before the next extended operating cycle.

Overall condition: AMBER — Maintenance action recommended.

2. Inspection Objective

The objective was to assess the operating condition of Unit 17, compare measured parameters with approved demonstration limits, identify deviations, and recommend appropriate maintenance actions.

3. Equipment Overview

Attribute	Recorded Value
Model	HT-500
Equipment Type	Francis hydraulic turbine
Unit	17
Rated Output	25 MW
Operating Hours	18,420 h
Last Major Service	12 Feb 2026
Inspection Team	Mechanical Maintenance — Demo Cell

Important: All equipment names, readings, limits and recommendations in this report are fictional and created for software demonstration purposes.

4. Inspection Conditions & Method

Measurements were recorded after the turbine reached a stable loaded operating condition. Instrument readings were captured using the facility's fictional digital maintenance checklist. The values below are representative synthetic data intended to exercise document extraction, comparison, calculation and decision-support functions.

Parameter	Observed	Recommended Range	Status
Bearing Temperature	87 °C	60–80 °C	HIGH
Hydraulic Pressure	11.2 bar	8.0–10.0 bar	HIGH
Shaft Vibration	7.1 mm/s	< 5.0 mm/s	HIGH
Lubricating Oil Level	72 %	80–100 %	LOW
Cooling Water Flow	118 L/min	100–130 L/min	NORMAL
Generator Load	82 %	70–90 %	NORMAL

5. Detailed Observations

5.1 Bearing Temperature: The recorded temperature of 87 °C is 7 °C above the upper recommended limit. The increase may be associated with lubrication condition, bearing friction, cooling effectiveness or load-related thermal effects. Further inspection is recommended.

5.2 Hydraulic Pressure: Pressure was recorded at 11.2 bar against a recommended upper limit of 10.0 bar. The deviation is 1.2 bar, or 12% above the upper limit. Pressure-control components and instrumentation should be checked.

5.3 Shaft Vibration: Vibration of 7.1 mm/s exceeds the recommended threshold of 5.0 mm/s. The reading should be investigated for alignment, imbalance, bearing condition and mechanical looseness.

5.4 Lubricating Oil: Oil level was 72%, below the preferred minimum of 80%. The level should be checked for leakage and restored according to the applicable maintenance procedure.

6. Parameter Deviation Summary

Parameter	Deviation	Interpretation
Bearing Temperature	7 °C above limit	Thermal condition requires inspection
Hydraulic Pressure	12% above limit	Pressure system review recommended
Shaft Vibration	2.1 mm/s above limit	Mechanical condition review recommended
Oil Level	8 percentage points below minimum	Lubrication check/top-up required

7. Maintenance Assessment

Based on the recorded readings, Unit 17 should remain under controlled observation until the identified deviations are investigated. The combined presence of elevated temperature and vibration increases the priority of mechanical inspection.

Finding	Priority	Recommended Action	Verification
Bearing temperature high	High	Inspect bearing and lubrication circuit; verify cooling	Repeat temperature measurement
Hydraulic pressure high	Medium	Inspect pressure-control system and sensor calibration	Record stable pressure
Vibration high	High	Check alignment, imbalance, bearing condition and fasteners	Repeat vibration test
Oil level low	Medium	Check for leakage and restore level per SOP	Confirm level after operation

8. Proposed Maintenance Sequence

Step	Activity	Expected Output
1	Place Unit 17 under controlled operating observation	No uncontrolled parameter escalation
2	Inspect bearing housing and lubrication circuit	Inspection findings recorded
3	Verify pressure instrumentation and control components	Calibrated/verified pressure reading
4	Inspect alignment, vibration sources and mechanical fasteners	Root-cause candidates identified
5	Check oil level and leakage indicators	Oil level restored within approved range
6	Run controlled post-maintenance test	All critical readings within limits
7	Engineering review and approval	Release / hold decision

9. Risk Assessment

If the deviations are ignored, continued operation could increase wear on mechanical components and reduce equipment reliability. The recommended approach is therefore to investigate the high-priority findings before an extended operating cycle.

Operational recommendation: Do not treat this report as authorization for maintenance or equipment release. Final action must be approved by an authorized engineering representative.

10. Internal SOP Cross-Check — Demonstration Extract

The following fictional rules represent the type of private organizational knowledge that a local RAG system could retrieve when evaluating this report.

SOP Rule	Requirement	Report Match
SOP-MECH-04	Bearing temperature above 80 °C requires inspection before extended operation.	87 °C — TRIGGERED
SOP-HYD-02	Hydraulic pressure should remain between 8.0 and 10.0 bar during loaded operation.	11.2 bar — TRIGGERED
SOP-VIB-03	Vibration above 5.0 mm/s requires mechanical condition assessment.	7.1 mm/s — TRIGGERED
SOP-LUB-01	Lubricating oil should remain at or above 80% unless otherwise approved.	72% — TRIGGERED
SOP-REL-05	Equipment release requires documented post-maintenance verification.	Pending.

11. Preliminary Recommendation

Based on the fictional inspection data and SOP cross-check, the recommended preliminary action is to initiate a maintenance inspection focused on the bearing/lubrication system, vibration sources and hydraulic pressure-control system. Following corrective action, Unit 17 should undergo a controlled verification run and documented engineering review.

12. Suggested Approval Note

“Inspection of Hydraulic Turbine HT-500 Unit 17 has identified elevated bearing temperature, hydraulic pressure and shaft vibration, together with low lubricating-oil level. The findings exceed the applicable demonstration limits. Maintenance inspection and corrective action are recommended before extended operation. Release of the unit should be subject to successful post-maintenance verification and authorized engineering approval.”

13. Data & Evidence Register

Evidence Item	Source	Use
Operating readings	Inspection Report HT500-IR-017-2026	Parameter comparison
Maintenance thresholds	Internal SOP demonstration extract	Grounding
Deviation calculations	Derived from observed vs. limit values	Decision support
Recommended actions	Assessment based on report + SOP	Draft deliverable

14. Review & Verification Record

This page is designed as the final verification gate in the demonstration workflow. It illustrates how an AI-generated deliverable can be reviewed before release.

Verification Check	Result	Reviewer Action
Observed values extracted correctly	PASS	Confirm against source report
Recommended limits extracted correctly	PASS	Confirm against SOP
Deviation calculations consistent	PASS	Spot-check calculations
Recommendations grounded in evidence	PASS	Review cited findings
Policy / SOP compliance	PASS	Confirm applicable procedure
Final engineering approval	PENDING	Authorized expert must approve

15. Final Status

STATUS: PENDING HUMAN APPROVAL

The report is suitable as a draft decision-support document. It must not be interpreted as an operational authorization, safety certification, or engineering release without review by an appropriately authorized human expert.

16. Demonstration Workflow

Confidential Report → Local Document Extraction → Private SOP Retrieval → Agentic Analysis → Controlled Calculation → Draft Approval Note → Validation → Human Approval

Prepared By	AI Workbench Demonstration
Reviewed By	_____
Engineering Approval	_____
Date	_____

END OF REPORT

SYNTHETIC DATA NOTICE: This document is entirely fictional. It does not describe a real turbine, facility, organization, maintenance event, operating limit, safety procedure or engineering recommendation.